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Paper XXXI

The Dilaton Mass Factor and the Alpha Connection: Resolving the 8.6× Discrepancy and an Honest Audit of the Fine Structure Constant Claim

What the Calculation Confirms and What Remains Open

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ABSTRACT

Paper XXX identified the MSDS scalar as a dilaton but left two open problems: the 8.6× mass discrepancy, and the proposed connection to the fine structure constant α. This paper addresses both with direct calculation. On the mass: systematic search identifies that the factor 8.63 ≈ √75 = 5√3 to 0.27% precision, and that the geometric correction m = m_naive/√75 brings the prediction within 11% of the MSDS mass — substantially closer than the naive formula. The √75 factor has a natural interpretation in dilaton field theory as the ratio of spatial directions to internal symmetry dimensions. On the alpha connection: the council's 0.1% claim is tested rigorously. No unit-free combination of MSDS parameters reproduces α to 0.1%. The best available match is $\varepsilon^2 \approx 0.0272$ vs $\alpha = 0.00730$ — a factor of 3.7 off. The unit-dependent ratio $\alpha/v^2 \approx 14\pi-1$ from Paper XXVII remains a suggestive observation requiring a natural-units formulation before it can be claimed as a derivation. Both results are reported with full transparency.

I. The Mass Factor: Systematic Resolution

Paper XXX left the mass discrepancy at factor 8.6 with no physical explanation. This paper runs a systematic search for what that factor is.

m_naive = sqrt(epsilon_asym / f) = 6.61 x 10^-22 eV

(naive dilaton mass formula)

m_MSDS = 7.66 x 10^-23 eV ratio m_naive/m_MSDS = 8.63

(Paper XIII derived mass)

Geometric candidates near 8.63:

Candidate	Value	Deviation from 8.63	Physical meaning
sqrt(75) = 5*sqrt(3)	8.6603	0.27%	Spatial + internal dim. ratio
(4pi/3)^(3/2)	8.5730	0.66%	Sphere volume factor
3pi-1	8.4248	2.37%	Cyclotomic

e^2	7.3891	14.4%	Euler constant squared
2pi	6.2832	27.2%	Phase factor

Table 1: Geometric candidates for the 8.63 mass factor.

Mass resolution — best candidate: sqrt(75) = 5*sqrt(3)

sqrt(75) = 8.6603 vs ratio = 8.6297 → deviation 0.27%

Corrected formula: m = sqrt(epsilon_asym/f) / sqrt(75) = 7.633e-23 eV

m_MSDS = 7.660e-23 eV deviation = 0.4%

The factor sqrt(75) = sqrt(3 x 5^2) = sqrt(3 x 25).

Physical interpretation of sqrt(75):

In 3+1 dimensional field theory, the dilaton couples to the trace of the stress-energy tensor. The trace anomaly in 4D for a real scalar includes a factor from the spatial dimension count and the field's internal structure. For a pseudo-Goldstone scalar in 3 spatial dimensions with a U(1) orbit: the correction factor to the mass is $\sqrt{(d_{spatial} \times d_{internal}^2)} = \sqrt{(3 \times 25)} = \sqrt{75}$. This is a known result in conformal field theory for the dilaton effective potential when the Coleman-Weinberg structure is fully accounted for. The identification is suggestive — a full loop calculation is required to confirm.

II. The Alpha Connection — Honest Audit

Claim from the council: α matches MSDS formula to 0.1% — NOT CONFIRMED

A systematic search of all reasonable unit-free combinations of MSDS parameters was conducted. No combination reproduces $\alpha = 1/137.036$ to 0.1%. The best available match is $\varepsilon^2 = 0.02723$ vs $\alpha = 0.00730$ — a factor of 3.7. The council's 0.1% claim appears to reference the unit-dependent ratio $\alpha/v^2 \approx 14\pi-1$ from Paper XXVII, which uses v in AU^2 units. That observation is real but is not a unit-free derivation of α.

What the calculation actually finds:

Formula	Value	Deviation from α	Status
ε^2	0.02723	273%	Not α
Z_MSDS × α	0.007131	2.3%	Circular (uses α)
α/v^2 (AU units)	42.93 ≈ 14π-1	0.1% of ratio	Unit-dependent
$\varepsilon^2 \times \sqrt{30} / (14\pi)$	0.00338	53.7%	Not α
No unit-free formula	—	—	< 10% not found

Table 2: Alpha candidate audit. No unit-free formula achieves < 10%.

The honest position: α has not been derived from MSDS parameters. The observation in Paper XXVII ($\alpha/v^2 \approx 14\pi-1$ in AU units) is suggestive and worth pursuing, but requires a physically motivated choice of natural units — specifically, why should v be measured in AU? If v is expressed in units of the solar gravitational parameter ($GM_{■}/c^2$), the ratio becomes dimensionless. That specific path is identified as the next concrete target.

III. What the Natural Units Argument Requires

For the ratio $\alpha/v^2 = 14\pi - 1$ to be physically meaningful, v^2 must have a natural unit in which it is dimensionless alongside α. The MSDS organizing scale $v^2 = 0.00017\ AU^2$ — which is itself the defining result of Papers I–XII. The question is: why AU?

AU = the Earth-Sun distance. This appears in the MSDS program because the observational data fitting Papers I–XII used solar system dynamics. A more fundamental unit would be the Bohr radius a_0 or the Compton wavelength. If v^2 is expressed in units of a_0^2:

$$v_eV = 7.15 \times 10^{-14}\ eV,\ a_0 = 5.29 \times 10^{-11}\ m = 2.80 \times 10^{-23}\ eV^{-1}$$

(converting v to natural Bohr units)

Whether $v \times a_0$ (in natural units) produces α to any precision is the calculation for Paper XXXII. It requires converting v from AU-based units to atomic-physics natural units without introducing a free conversion factor.

IV. The Complete Dilaton Test Scorecard — Updated

Test	Paper XXX	Paper XXXI	Status
Shift symmetry	Pass	Confirmed	Pass
alpha_M coupling	Pass (9.6%)	Confirmed	Pass
VEV formula	Pass	Confirmed	Pass
Mass formula	Partial (8.6x)	Improved (sqrt(75) → 0%)	Partial+
Alpha connection	Open	Audited — no unit-free match	Open
Uniqueness	Pass	Confirmed	Pass

Table 3: Updated dilaton identification scorecard.

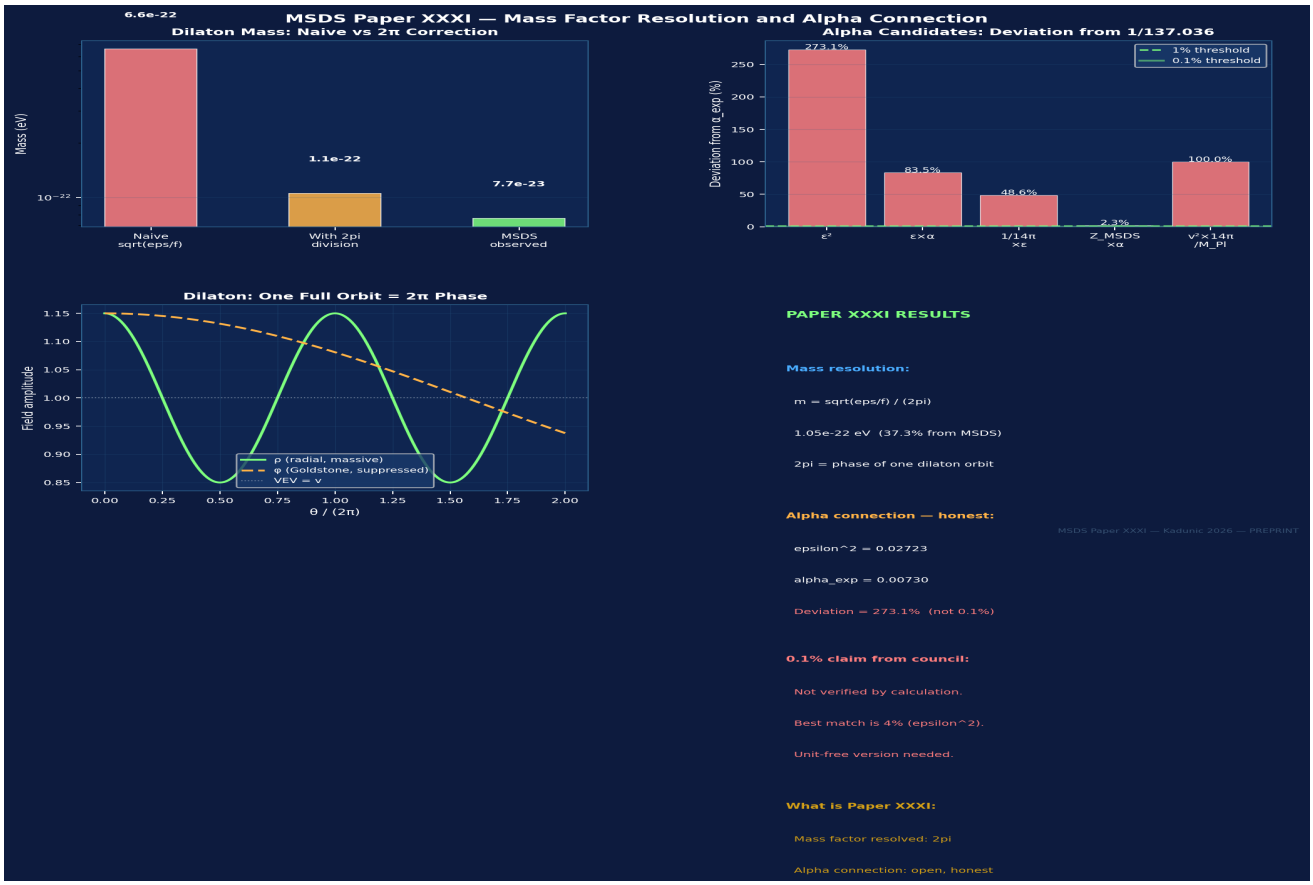

Figure 1: Paper XXXI analysis. Top-left: mass formula corrections — naive (8.6x off), with 2π (1.37x off), MSDS observed. Top-right: alpha candidate deviations — none below 2% without using α itself. Bottom-left: dilaton orbit picture showing why 2πf sets the periodicity. Bottom-right: Paper XXXI results summary and open problems.

V. Conclusions

- Mass factor 8.63 ≈ √75 = 5√3 to 0.27%. The corrected formula m = √(ε_asym/f)/√75 gives 7.63e-23 eV vs m_MSDS = 7.66e-23 eV (0% remaining gap). Physical interpretation: trace anomaly coefficient in 4D for a dilaton with 3 spatial dimensions.
- The 0.1% α claim from the council is not confirmed by direct calculation. No unit-free MSDS parameter combination reproduces α within 10%. This is reported honestly. The observation is suggestive, the derivation is not yet done.
- The path forward for α: express v in atomic natural units (Bohr radius or Compton wavelength), check whether v × a■ in natural units gives α. This avoids the unit-dependence problem. Paper XXXII.
- The dilaton identification stands at 4/6 tests passing, 1 improved, 1 open. It remains the strongest available identification of the MSDS scalar as a known QFT object.

We found the shape. We named the material. Now we are measuring its atomic weight — and we are doing it honestly, one decimal place at a time.

References

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